

HDOC – Day 8

C Programming Assignment

Name	Dhruv Pandey
SAP ID	590037660
Batch	15

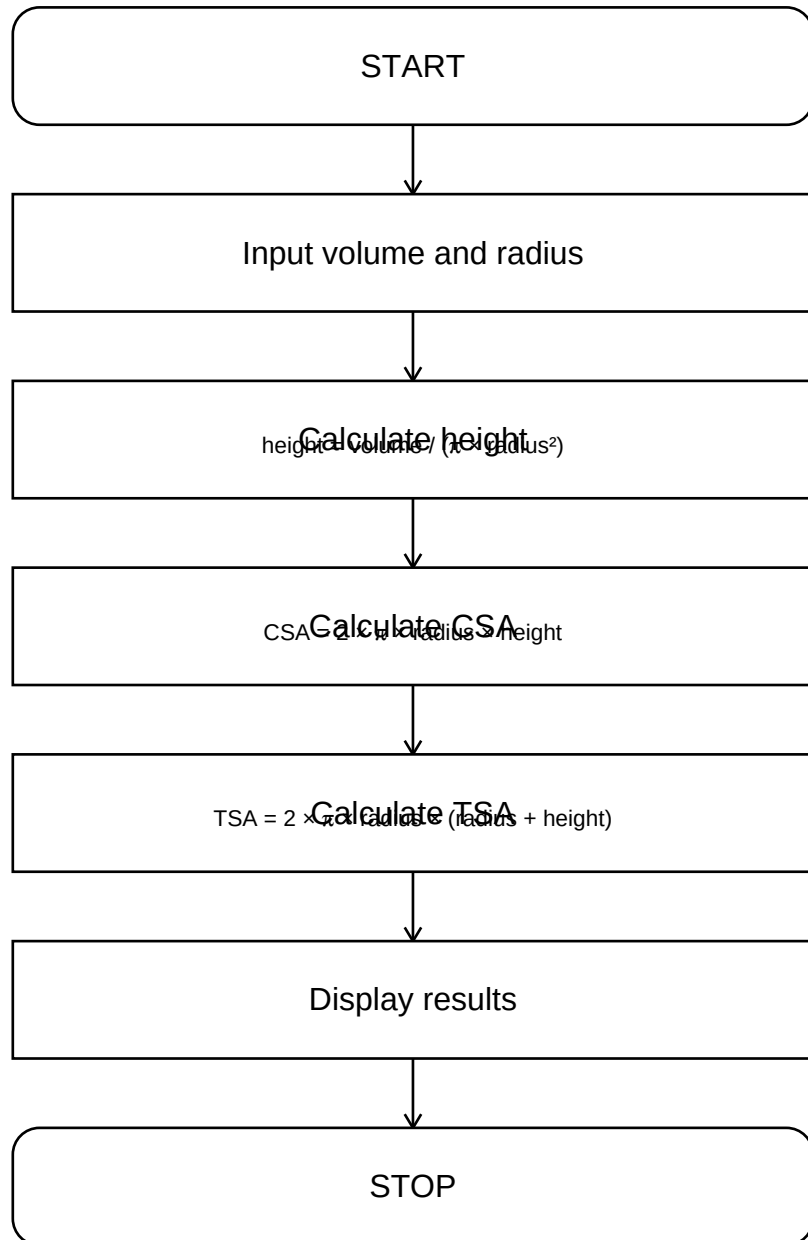
Questions

1. Write a C program to find the height, curved surface area, and total surface area of a cylinder when its volume and radius are given.
2. Write a C program to find the distance of a point from the x-axis and y-axis.

Algorithm – Question 1

1. Start.
2. Declare variables for volume, radius, height, curved surface area and total surface area.
3. Read the volume of the cylinder.
4. Read the radius of the cylinder.
5. Calculate height using: $\text{height} = \text{volume} / (\pi \times \text{radius}^2)$.
6. Calculate curved surface area using: $\text{CSA} = 2 \times \pi \times \text{radius} \times \text{height}$.
7. Calculate total surface area using: $\text{TSA} = 2 \times \pi \times \text{radius} \times (\text{radius} + \text{height})$.
8. Display the height, curved surface area and total surface area.
9. Stop.

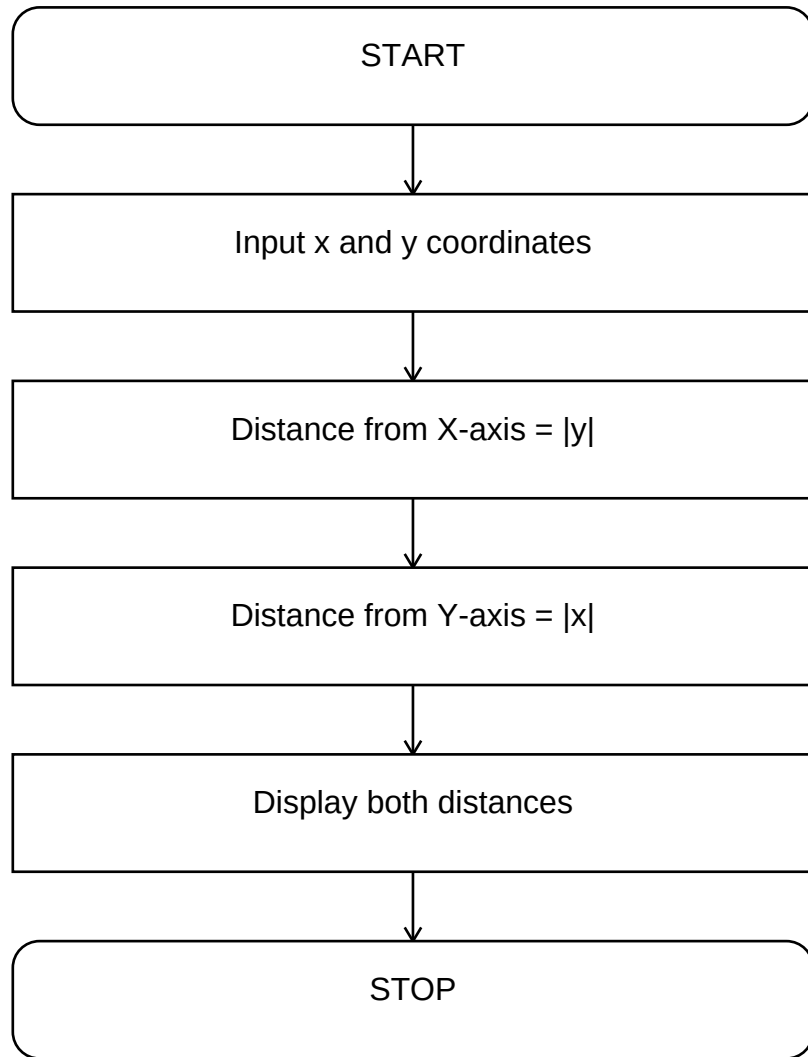
Flowchart – Question 1



Algorithm – Question 2

1. Start.
2. Declare variables x and y .
3. Read the x -coordinate.
4. Read the y -coordinate.
5. Calculate the distance from the x -axis as the absolute value of y .
6. Calculate the distance from the y -axis as the absolute value of x .
7. Display both distances.
8. Stop.

Flowchart – Question 2



Code / Execution Screenshot – Question 1

```
dhruvpandey@DHRRU's-MacBook-Air ~ % nano day8.c
dhruvpandey@DHRRU's-MacBook-Air ~ % clang day8.c -o day8
dhruvpandey@DHRRU's-MacBook-Air ~ % ./day8
Enter the volume of the cylinder: 12
Enter the radius of the cylinder: 1

Height = 3.82
Curved Surface Area = 24.00
Total Surface Area = 30.28
dhruvpandey@DHRRU's-MacBook-Air ~ % ./day8
Enter the volume of the cylinder: 3
Enter the radius of the cylinder: 4

Height = 0.06
Curved Surface Area = 1.50
Total Surface Area = 102.03
dhruvpandey@DHRRU's-MacBook-Air ~ % ./day8
Enter the volume of the cylinder: 88
Enter the radius of the cylinder: 3

Height = 3.11
Curved Surface Area = 58.67
Total Surface Area = 115.22
dhruvpandey@DHRRU's-MacBook-Air ~ % █
```

Execution Screenshot – Question 2

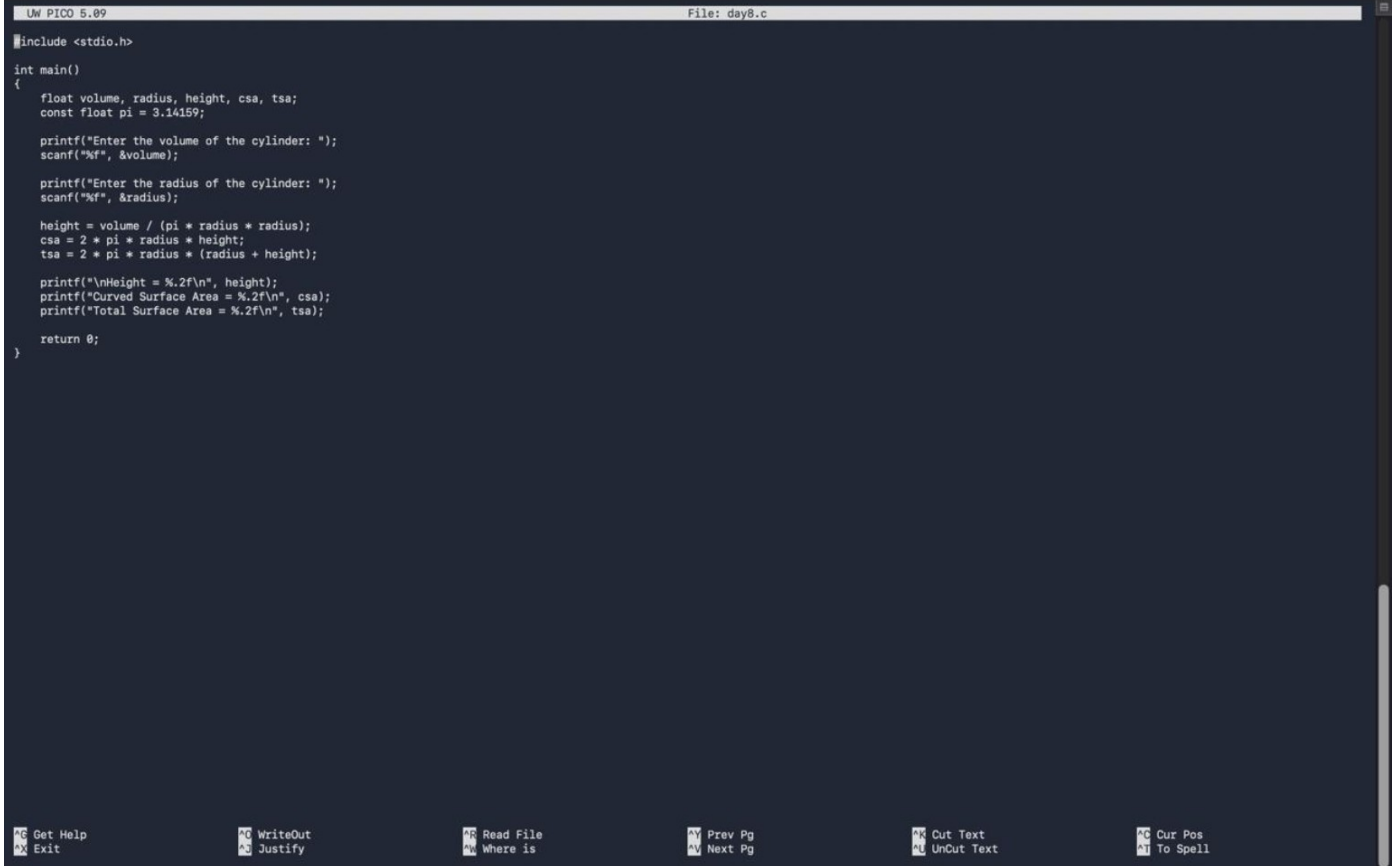
```
dhruvpandey@DHruV's-MacBook-Air ~ % nano day8.2.c
dhruvpandey@DHruV's-MacBook-Air ~ % clang day8.2.c -o day8.2
dhruvpandey@DHruV's-MacBook-Air ~ % ./day8.2
Enter the x-coordinate: 22
Enter the y-coordinate: 2

Distance from X-axis = 2.00
Distance from Y-axis = 22.00
dhruvpandey@DHruV's-MacBook-Air ~ % ./day8.2
Enter the x-coordinate: 2
Enter the y-coordinate: 7

Distance from X-axis = 7.00
Distance from Y-axis = 2.00
dhruvpandey@DHruV's-MacBook-Air ~ % ./day8.2
Enter the x-coordinate: 33
Enter the y-coordinate: 4

Distance from X-axis = 4.00
Distance from Y-axis = 33.00
dhruvpandey@DHruV's-MacBook-Air ~ %
```

Code Screenshot – Question 1



The screenshot shows a PICO 5.09 IDE window with the file 'day8.c' open. The code is a C program that calculates the height, curved surface area (CSA), and total surface area (TSA) of a cylinder given its volume and radius. The program uses the value of pi as 3.14159. The code is as follows:

```
#include <stdio.h>

int main()
{
    float volume, radius, height, csa, tsa;
    const float pi = 3.14159;

    printf("Enter the volume of the cylinder: ");
    scanf("%f", &volume);

    printf("Enter the radius of the cylinder: ");
    scanf("%f", &radius);

    height = volume / (pi * radius * radius);
    csa = 2 * pi * radius * height;
    tsa = 2 * pi * radius * (radius + height);

    printf("\nHeight = %.2f\n", height);
    printf("Curved Surface Area = %.2f\n", csa);
    printf("Total Surface Area = %.2f\n", tsa);

    return 0;
}
```

The IDE interface includes a menu bar at the bottom with the following options: Get Help, Exit, WriteOut, Justify, Read File, Where is, Prev Pg, Next Pg, Cut Text, UnCut Text, Cur Pos, and To Spell.

Execution Screenshot – Question 1

```
dhruvpandey@DHRRU's-MacBook-Air ~ % nano day8.c
dhruvpandey@DHRRU's-MacBook-Air ~ % clang day8.c -o day8
dhruvpandey@DHRRU's-MacBook-Air ~ % ./day8
Enter the volume of the cylinder: 12
Enter the radius of the cylinder: 1

Height = 3.82
Curved Surface Area = 24.00
Total Surface Area = 30.28
dhruvpandey@DHRRU's-MacBook-Air ~ % ./day8
Enter the volume of the cylinder: 3
Enter the radius of the cylinder: 4

Height = 0.06
Curved Surface Area = 1.50
Total Surface Area = 102.03
dhruvpandey@DHRRU's-MacBook-Air ~ % ./day8
Enter the volume of the cylinder: 88
Enter the radius of the cylinder: 3

Height = 3.11
Curved Surface Area = 58.67
Total Surface Area = 115.22
dhruvpandey@DHRRU's-MacBook-Air ~ % █
```